

Pranav Dhiran

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PROFILE

AI Research Intern at AIISC (University of South Carolina) working on neurosymbolic SLM architecture, custom pre-training pipelines, and GRPO-based RL fine-tuning. LFX '26 Mentee at Hyperledger Cello building an AI agent over production REST APIs; active CNCF contributor to Meshery. Spans formal methods (Z3, Lean 4), transformer pre-training, multi-agent systems, and open-source LLM-hardware integrations — looking to work on large codebases where there is real engineering to learn from.

EDUCATION

B.Tech — Electronics & Telecommunication Engineering 2023 – Present
SGGS Institute of Engineering & Technology, Nanded | Minor in IT

EXPERIENCE

LFX Mentee Jun 2026 – Present
Hyperledger Cello · Linux Foundation

- Hyperledger Cello manages the full lifecycle of Hyperledger Fabric networks — every operation today requires manual dashboard interaction, creating a steep learning curve.
- Building an AI agent that takes natural language input, reasons over the right sequence of Cello API calls, and executes them — turning network operations and Chaincode debugging into a conversation.

Research Intern Apr 2026 – Present
AIISC, University of South Carolina

- Designing neurosymbolic SLM architecture integrating symbolic reasoning constraints into pre-training — custom BPE tokenizer with AGROVOC entity injection, Z3-based causal KG verification, and Triple Transformer Encoder.
- GRPO-based RL fine-tuning for SLM alignment — reward modeling via self-certainty scores, policy optimization, and evaluation on neurosymbolic reasoning benchmarks.

Open Source Contributor Mar 2026 – Present
Meshery — CNCF Sandbox Project

- 5+ merged PRs — service mesh management features, UI components, and API integrations across Go backend and React frontend.
- Active in code reviews, issue triage, and community discussions per CNCF contributor guidelines.

PROJECTS

GNU Radio MCP Server — LLM-to-SDR Bridge — Python · FastMCP · ZMQ · XML-RPC · GNU Radio · PyPI · GitHub

- Built and published a provider-agnostic MCP server to PyPI bridging LLMs to live GNU Radio SDR flowgraphs via stdio transport — 13 tools covering flowgraph lifecycle, frequency/gain control, and signal monitoring; Pydantic v2 validation throughout.
- ZMQ IQ streaming with async capture and Welch PSD computation for real-time spectrum data; XML-RPC control plane for GNU Radio 3.10+; enables natural-language control of HackRF One and RTL-SDR hardware directly through Claude Desktop.

Small Language Model from Scratch — PyTorch · Hugging Face · AMP · NLP · GitHub

- Pre-trained a GPT-style autoregressive transformer (6L, 6H, 384-dim) on TinyStories — custom BPE tokenization, mmap pipelines, AMP mixed precision, gradient accumulation, cosine LR; systematic depth/embedding ablations for scaling behavior analysis.

Medaura — Agentic Pharmacy System — FastAPI · LangChain · ChromaDB · Langfuse · React · Live

- 5-agent autonomous pharmacy system (Ordering, Safety, Forecast, Procurement, UI) with semantic RAG via ChromaDB + sentence-transformers and Langfuse observability — live on web, zero human intervention, 4 languages.

RF Watch — Open-Source RF Spectrum Monitor — Python · GNU Radio · HackRF One · Signal Processing · GitHub

- Real-time RF spectrum analyzer — FFT-based spectral feature extraction with lightweight ML classifier for passive transmitter detection; born from SIH 2025 anti-drone work for ITBP.

TECHNICAL SKILLS

Languages & Frameworks:	Python, Go (learning), PyTorch, TensorFlow, Hugging Face Transformers, TRL
APIs & Protocols:	REST, gRPC, GraphQL, OpenAI/Gemini/Ollama APIs, FastMCP, XML-RPC, ZMQ
LLM Engineering:	Instruction Fine-Tuning, LoRA, PEFT, GRPO, RLHF, INT4/INT8 Quantization, Unsloth, LangChain, LangGraph, ChromaDB, FAISS
Agentic & Infra:	Multi-Agent Systems, Tool-Use, Function Calling, MCP Servers, LangSmith, Langfuse, Docker (basic), Kubernetes (basic)

CERTIFICATIONS

Fine-tuning & RL for LLMs: Intro to Post-training — DeepLearning.AI · AI Agents in LangGraph — DeepLearning.AI · Quantization Fundamentals — Hugging Face · MCP: Build Rich-Context AI Apps — Anthropic

AWARDS & RECOGNITION

- International Finalist (Top 6) — UWA Hack for Impact 2026
- National Finalist — Smart India Hackathon (SIH) 2024 & 2025
- Regional Qualifier — Nxt Wave × OpenAI Buildathon